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Information Sharing and Rating Manipulation

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1 Big picture

- **Question.** Does public information sharing through a credit registry improve credit allocation, or do banks manipulate the information before disclosure?
- **Setting.** Argentina expanded its public credit registry in 1998 after CD-ROM distribution made it cheap to disclose information for small borrowers previously below the disclosure threshold.
- **Unique timing.** The reform was announced in April 1998 and implemented in July 1998. The interim period lets the authors observe ratings after banks learned they would become public but before other banks could learn from them.
- **Main manipulation result.** Banks strategically downgraded high-quality borrowers with single lending relationships before the ratings were disclosed, especially when lenders had stronger informational rents.
- **Second manipulation result.** Banks upgraded low-quality borrowers with multiple lenders before disclosure, consistent with avoiding creditor runs.
- **Why this matters.** Public credit registries are supposed to reduce adverse selection. If banks supply manipulated ratings, the registry may disclose less informative information than policymakers expect.
- **Effect on credit allocation.** The registry still improves access for some borrowers, but strategically downgraded borrowers do not obtain the same benefits.
- **Policy takeaway.** Information disclosure can backfire when the disclosed agent controls the information and disclosure changes that agent's payoff.

2 Data and measurement

2.1 Argentinian public credit registry reform

- Before the reform, the Central Bank collected credit information from banks but publicly disseminated ratings mainly for borrowers with total debt above USD 200,000 and borrowers in default.
- In April 1998, the Central Bank announced that CD-ROM distribution would make it possible to disclose ratings for roughly 540,000 smaller borrowers below the threshold.
- The expansion was implemented in July 1998.
- This creates three periods: pre-announcement January–March 1998, interim April–June 1998, and post-expansion from July 1998 onward.

Interpretation. The interim period is the key window: banks know disclosure is coming, but the market has not yet received the ratings.

2.2 Treatment and control samples

- **Treatment borrowers:** total borrowing between USD 150,000 and USD 200,000 in the pre-announcement period.
- **Control borrowers:** total borrowing between USD 200,000 and USD 250,000.
- The narrow window around the USD 200,000 threshold is designed to make treatment and control borrowers comparable.
- The unit of observation is usually borrower-bank-month.

Interpretation. The design resembles a local difference-in-differences around a disclosure threshold.

2.3 Credit ratings, relationship types, and outcomes

- A rating of 1 denotes good standing; ratings above 1 represent increasing degrees of repayment problems.
- A downgrade from 1 to 2 is the main signal of strategic concealment for initially high-quality borrowers.
- An upgrade from 2 to 1 is the main signal of strategic protection for initially lower-quality borrowers with multiple lenders.
- Single-lender borrowers are more likely to be informationally captured by one bank.
- Multiple-lender borrowers are the natural group for studying creditor-run incentives.
- Main outcomes include rating changes, future default, log total debt, log number of banks, percent local lenders, and percent foreign lenders.

3 Models and empirical specifications

3.1 Baseline difference-in-differences for downgrades

$$Downgrade_{ibt} = \alpha_i + \tau_t + \beta(Treated_i \times Interim_t) + \Gamma Controls_{ibt} + \varepsilon_{ibt}.$$

- *Downgrade* equals one if a bank changes a borrower rating from 1 to 2.
- *Treated* equals one for borrowers below the USD 200,000 threshold but within the local sample window.
- *Interim* equals one during April–June 1998.

Interpretation. If banks strategically conceal good borrowers, β should be positive for high-quality single-lender borrowers.

3.2 Local-bank heterogeneity

$$Downgrade_{ibt} = \dots + \beta(Treated_i \times Interim_t) + \theta(Treated_i \times Interim_t \times Local_b) + \varepsilon_{ibt}.$$

Interpretation. A positive θ means the interim downgrading effect is larger for local banks, which are expected to have stronger private information and relationship rents.

3.3 Multiple-lender upgrade specification

$$Upgrade_{ibt} = \alpha_i + \tau_t + \beta(Treated_i \times Interim_t) + \Gamma Controls_{ibt} + \varepsilon_{ibt}.$$

Interpretation. Banks may inflate low-quality multiple-lender borrowers to prevent other lenders from withdrawing credit.

3.4 Rating informativeness and relationship outcomes

$$Default_{i,t+h} = \alpha_i + \tau_t + \beta(Treated_i \times Pre_t) + \Gamma Controls_i + \varepsilon_i.$$

$$Y_{it} = \alpha_i + \tau_t + \beta(Treated_i \times Post_t) + \Gamma Controls_i + \varepsilon_{it}.$$

Interpretation. Manipulation changes the information content of ratings and affects whether public disclosure improves credit access.

4 Identification and empirical design

- The reform changes disclosure status for borrowers below the USD 200,000 threshold but not for similar borrowers just above it.
- The interim period separates manipulation from learning: banks can alter ratings before competitors see them.
- Treatment and control borrowers are selected from a narrow window around the threshold.
- Figures and descriptive statistics show similar distributions by industry, size, geography, and borrower characteristics.
- Effects are strongest where theory predicts: local banks, exclusive relationships, more opaque borrowers, and competitive banking markets.

Interpretation. The strongest evidence is the combination of timing, local threshold design, and mechanism-consistent cross-sectional tests.

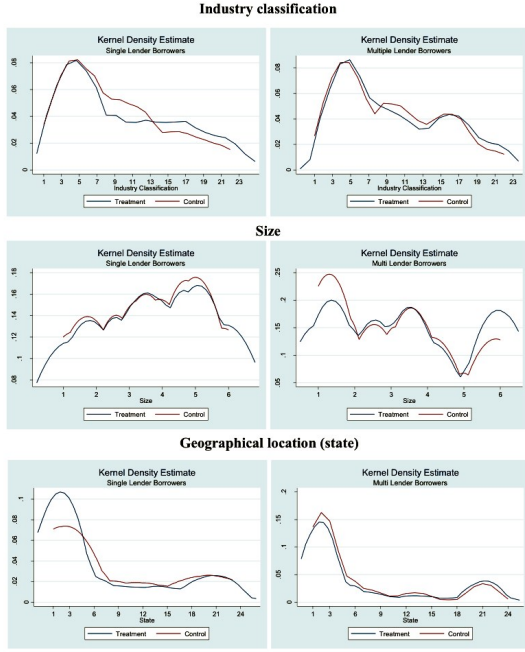
5 Tables and figures

The figures and tables below are directly cropped from the original paper and grouped to save space.

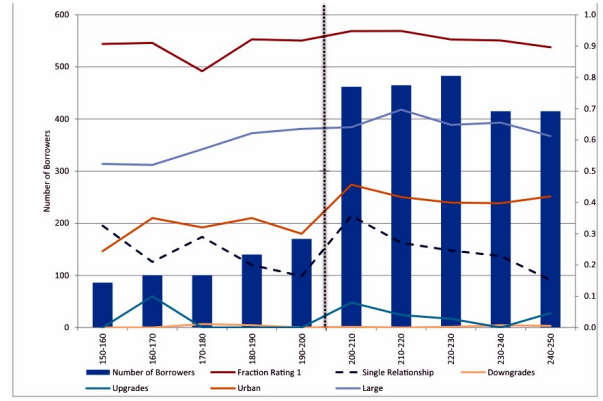
5.1 Table 1 and Figures 1–2: sample balance and threshold validity

Read: Figure 1 shows comparable treatment and control distributions by industry, size, and state. Figure 2 shows smoothness around the USD 200,000 threshold. Table 1 shows treatment and control borrowers are broadly similar aside from debt size.

Economic message. The threshold design is credible because treated and control borrowers look similar before the disclosure shock.



(a) Figure 1: observable distributions



(b) Figure 2: borrowers around threshold

5.2 Figures 3–5 and Table 2: strategic downgrades

Read: Single-lender treated borrowers experience abnormal downgrades in the interim period. The effect is stronger for local banks and visible in monthly dynamics.

Economic message. Banks with private information appear to downgrade good exclusive borrowers before their ratings become public.

Table 1
Descriptive statistics

A. Characteristics of single and multiple bank-relationships

	All			Treatment			Control		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
	n = 8,522			All borrowers n = 1,791			n = 6,731		
Loan amount	130,534	151,450	83,864	113,167	133,000	67,129	135,155	161,300	87,204
Total debt	214,305	216,700	25,743	179,517	179,700	21,550	223,562	222,500	17,543
Collateral	0.63	1.00	0.48	0.66	1.00	0.47	0.62	1.00	0.49
Single relationship	0.25	0.00	0.43	0.22	0.00	0.42	0.25	0.00	0.43
Number of lenders	2.28	2.00	1.21	2.24	2.00	1.20	2.29	2.00	1.21
Credit rating	1.08	1.00	0.29	1.09	1.00	0.30	1.08	1.00	0.29
Size	3.42	3.00	1.88	3.19	3.00	1.84	3.49	3.00	1.88
Urban	0.40	0.00	0.49	0.31	0.00	0.46	0.42	0.00	0.49
	n = 2,103			Single relationship borrowers n = 402			n = 1,701		
Loan amount	211,759	213,000	24,137	176,147	177,600	16,903	220,175	219,350	16,798
Total debt	211,759	213,000	24,137	176,147	177,600	16,903	220,175	219,350	16,798
Collateral	0.73	1.00	0.45	0.69	1.00	0.46	0.74	1.00	0.44
Single relationship	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	0.00
Number of lenders	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	0.00
Credit rating	1.09	1.00	0.30	1.08	1.00	0.28	1.09	1.00	0.30
Size	3.71	4.00	1.85	3.67	4.00	1.83	3.72	4.00	1.86
Urban	0.46	0.00	0.50	0.38	0.00	0.49	0.48	0.00	0.50
	n = 6,419			Multiple relationship borrowers n = 1,389			n = 5,030		
Loan amount	103,923	97,700	79,226	94,940	97,100	65,174	106,404	97,950	82,521
Total debt	215,139	218,000	26,195	180,492	181,100	22,630	224,707	223,850	17,643
Collateral	0.60	1.00	0.49	0.65	1.00	0.48	0.58	1.00	0.49
Single relationship	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Number of lenders	2.70	2.00	1.11	2.59	2.00	1.13	2.73	2.00	1.10
Credit rating	1.08	1.00	0.29	1.09	1.00	0.31	1.08	1.00	0.29
Size	3.33	3.00	1.88	3.05	3.00	1.82	3.41	3.00	1.88
Urban	0.38	0.00	0.48	0.30	0.00	0.46	0.40	0.00	0.49

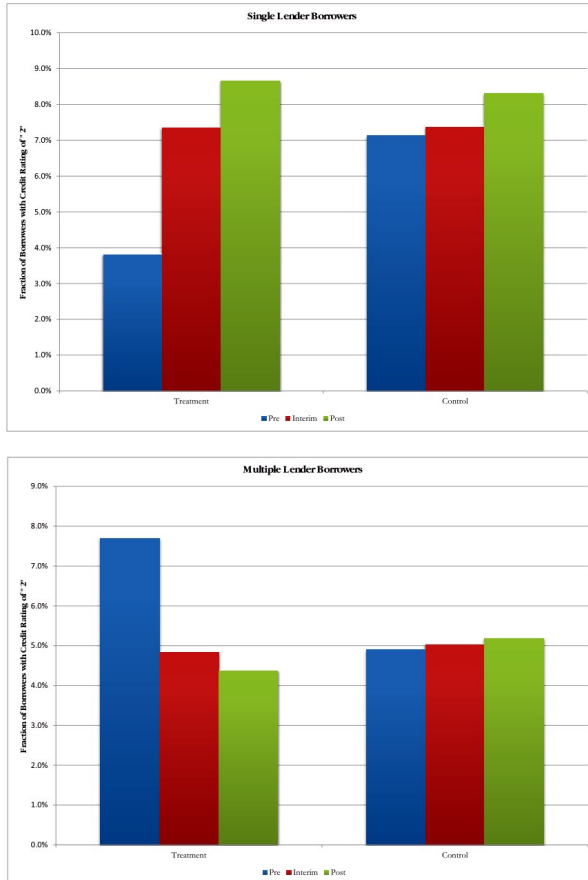
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Table 1
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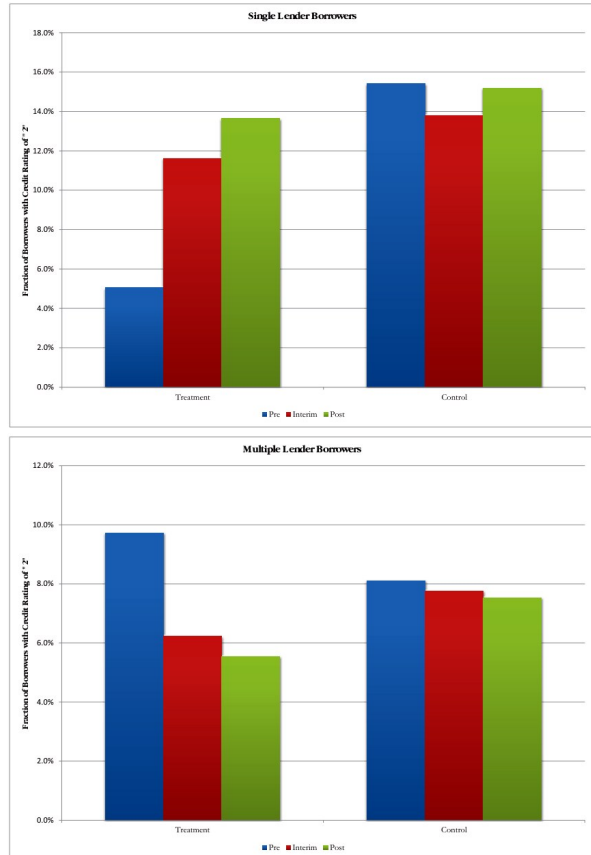
B. Bank relationships across sample periods

Period =	Sample means			
	All	Pre	Interim	Post
	All borrowers n = 46,891			
Loan amount	126,521	130,534	129,646	124,540
Total debt	221,312	214,305	221,520	223,234
Collateral	0.65	0.63	0.64	0.65
Single relationship	0.24	0.30	0.25	0.22
Number of lenders	2.61	2.28	2.58	2.72
Credit rating	1.24	1.08	1.17	1.30
Size	3.41	3.42	3.42	3.40
Urban	0.38	0.40	0.40	0.38
	Single relationship borrowers n = 11,698			
Loan amount	198,213	211,759	206,249	192,235
Total debt	221,834	211,759	212,983	227,066
Collateral	0.75	0.73	0.74	0.75
Single relationship	0.83	1.00	0.91	0.76
Number of lenders	1.23	1.00	1.13	1.33
Credit rating	1.32	1.09	1.19	1.42
Size	3.71	3.71	3.71	3.71
Urban	0.44	0.46	0.46	0.43
	Multiple relationship borrowers n = 35,193			
Loan amount	102,691	103,923	103,883	102,020
Total debt	221,138	215,139	224,391	221,959
Collateral	0.61	0.60	0.61	0.62
Single relationship	0.04	0.07	0.02	0.04
Number of lenders	3.07	2.70	3.07	3.18
Credit rating	1.21	1.08	1.16	1.26
Size	3.31	3.33	3.32	3.29
Urban	0.37	0.38	0.37	0.36

This table presents descriptive statistics for the borrower-bank relationships across treatment and control borrowers during the pre-announcement period only (panel A), and for the pre, interim, and post-credit reform expansion periods (panel B). Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The registry expansion announcement (the interim period) covers April to June 1998, and the post-expansion period, when information has been made public, includes the twelve months from July 1998 onward. Panels A and B are divided in three panels following different samples: for all borrower-bank relationships; for single relationship borrowers; and for multiple relationship borrowers. Loan amount is the dollar loan for each borrower-bank relationship; total debt is the total dollar amount of loans for each borrower across all bank relationships; collateral is an indicator variable equal to one if the loan is secured with collateral; single relationship is an indicator variable equal to one if the borrower has just one bank relationship; number of lenders is the number of lenders the borrower has; credit ratings are assigned by each lender to a borrower, and are integer between one (best) and five (worst); size is an indicator variable equal to one if the borrower size classification is equal or greater than three (>25 employees) and urban is an indicator variable equal to one if the borrower is located in the capital city of each



(a) Figure 3: rating changes around reform



(b) Figure 4: local-bank patterns

Table 2
Strategic downgrades of high-quality borrowers

Dependent variable	Downgrades from 1 to 2					
Sample lender	Single					3-month post
Period	All					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.004 (0.004)	−0.006 (0.007)	0.002 (0.004)	0.008 (0.007)	0.000 (0.004)	0.004 (0.005)
Treated × Interim	0.026** (0.012)	0.036*** (0.014)	−0.004 (0.009)	−0.011 (0.011)	0.002 (0.009)	−0.001 (0.009)
Treated × Post		0.013 (0.008)		−0.008 (0.008)		
Treated × Interim × Local			0.053** (0.021)	0.064*** (0.023)	0.038* (0.020)	0.054** (0.021)
Treated × Post × Local				0.034 (0.024)		
Bank controls	Yes	Yes	Yes	Yes	No	No
Borrower controls	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	No	No
Bank fixed effects	Yes	Yes	Yes	Yes	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	No	No
Bank-time fixed effects	No	No	No	No	Yes	Yes
Industry-time fixed effects	No	No	No	No	Yes	Yes
No. observations	9,558	9,558	9,558	9,558	9,558	5,151
R-sq.	0.03	0.03	0.03	0.03	0.16	0.18

This table presents the difference-in-differences effect of the credit registry expansion announcement (interim period) and public information (post-expansion period) on downgrade hazard rates, estimated using OLS. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The dependent variable, downgrade, is from rating = 1 to rating = 2. The sample is all treatment and control borrowers with a rating = 1 in the pre-announcement period, and with a single lending relationship in the pre-announcement period. Columns (1) to (5) use a 12-month post-period horizon, while Column (6) uses a three-month post-period horizon. We exclude borrowers that have been downgraded from the sample, explaining why the number of observations is lower than in Table 1. Local is an indicator variable equal to one if the lender is a local bank. The unit of observation is at the borrower-bank-month level. Bank controls include log(total assets), return on assets, capitalization (%), log(number of branches), nonperforming loans-to-total assets, deposits-to-total assets, and growth on total assets, while borrower controls

(a) Table 2: strategic downgrades

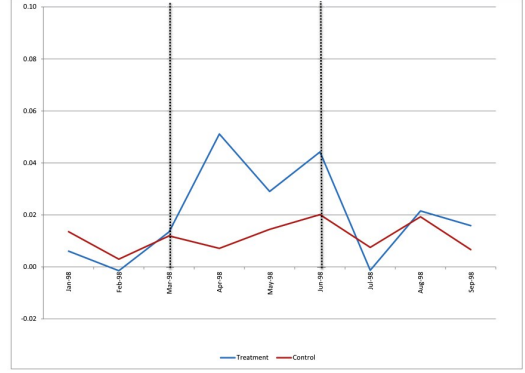


Figure 5
Conditional estimates of strategic downgrades by month for single lenders
This figure reports single relationship lenders' propensity to downgrade high-quality borrowers by month. The parameter estimates reported are for the coefficient of the interaction between the *Treated* dummy and monthly dummies for each month from January 1998 to September 1998. The approximate estimates are obtained by

(b) Figure 5: monthly dynamics

5.3 Tables 3–4: mechanism tests for downgrading

Read: Table 3 shows the effect is weaker when multiple lenders reduce the incumbent's information advantage. Table 4 shows weaker downgrading for large borrowers, urban borrowers, and borrowers in low-competition markets.

Economic message. Manipulation is concentrated where adverse selection and poaching risk are strongest.

Table 3
Cross-sectional lender differences in incentives to strategically downgrade

Dependent variable	Downgrades from 1 to 2			
	All			
	(1)	(2)	(3)	(4)
Treated	-0.000 (0.002)	0.001 (0.003)	-0.000 (0.003)	0.003 (0.004)
Treated $\times$ Interim	0.015*** (0.005)	0.001 (0.006)	0.003 (0.007)	0.016 (0.011)
Treated $\times$ Interim $\times$ Local		0.026** (0.011)	0.040*** (0.015)	0.022** (0.011)
Treated $\times$ Post			0.002 (0.004)	
Treated $\times$ Post $\times$ Local			0.015 (0.010)	
log(#banks)				-0.002 (0.002)
Treated $\times$ Interim $\times$ log(#banks)				-0.016* (0.009)
Borrower controls	Yes	Yes	Yes	Yes
Bank-time fixed effects	Yes	Yes	Yes	Yes
Industry-time fixed effects	Yes	Yes	Yes	Yes
No. observations	35,269	35,269	35,269	35,269
R-sq.	0.08	0.08	0.08	0.08

This table presents the difference-in-differences effect of the registry expansion announcement (interim period) and public information (post-expansion period) on downgrade hazard rates by bank-type, estimated using OLS. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The dependent variable, downgrade, is from rating = 1 to rating = 2. The sample is all treatment and control borrowers with a single and multiple lending relationships and with a rating of one in the pre-announcement period. The unit of observation is at the borrower-bank-month level. Local is an indicator variable equal to one if the lender is a local bank; log(#banks) is the log of the total number of bank-borrower relationships. We omit from reporting the coefficient estimates of *Treated $\times$ Local* in Columns (2), (3), and (4). Borrower controls include log(debt), state, size, industry classification, and urban. Standard errors are clustered at the borrower level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level.

(a) Table 3: lender differences

Table 4
Cross-sectional borrower differences in incentives to strategically downgrade

Dependent variable	Downgrades from 1 to 2					
Sample lenders	Single					
Characteristic	Large borrower		Urban borrower		Low competition	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.010 (0.007)	-0.009 (0.006)	0.002 (0.006)	-0.008 (0.006)	0.003 (0.004)	-0.000 (0.004)
Treated × Interim	0.058** (0.024)	-0.001 (0.012)	0.043** (0.018)	0.001 (0.011)	0.026** (0.012)	0.004 (0.009)
Treated × Interim × Characteristic	-0.048* (0.028)	0.003 (0.013)	-0.047** (0.021)	0.000 (0.017)	-0.030* (0.016)	-0.029* (0.016)
Treated × Interim × Local		0.077** (0.032)		0.056** (0.026)		0.035* (0.020)
Treated × Interim × Characteristic × Local		-0.066* (0.041)		-0.070* (0.037)		
Borrower controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank-time fixed effects	No	Yes	Yes	Yes	Yes	Yes
Industry-time fixed effects	No	Yes	Yes	Yes	Yes	Yes
No. observations	9,558	9,558	9,558	9,558	9,558	9,558
R-sq.	0.16	0.16	0.16	0.16	0.16	0.16

This table presents the difference-in-differences effect of the registry expansion announcement (interim period) on downgrade hazard rates by borrower-type, estimated using OLS. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The dependent variable, downgrade, is from rating = 1 to rating = 2. The sample is all treatment and control borrowers with a single lending relationship and with a rating of one in the pre-announcement period. The unit of observation is at the borrower-bank-month level. Characteristic denotes whether the borrower is a large borrower (Columns (1) and (2)) or urban borrower (Columns (3) and (4)), or whether the lender faces low competition (Columns (5) and (6)). Large borrowers is an indicator variable equal to one if the borrower size classification is equal or greater than three (>25 employees); urban is an indicator variable equal to one if the borrower is located in the capital city of each state; and local is an indicator variable equal to one if the lender is a local bank. Low competition is measured as the ratio of the total number of borrowers over the total number of branches for each lender at the state level. Local is an indicator variable equal to one if the lender is a local bank. We omit from reporting the coefficient estimates of *Treated $\times$ Characteristic*, *Interim $\times$ Characteristic* in all columns; *Treated $\times$ Local*, *Interim $\times$ Local*, *Treated $\times$ Local $\times$ Characteristic*, *Interim $\times$ Local $\times$ Characteristic* in Columns (2) and (4). Borrower controls include log(debt), state, and industry classification in all columns plus urban in Columns (1) and (2), log(debt) in Columns (3) and (4), and urban and log(debt) in Columns (5) and (6). Standard errors are clustered at the borrower level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level.

(b) Table 4: opacity and competition

5.4 Tables 5–7: rating informativeness and strategic upgrades

Read: Table 5 shows rating-2 borrowers become safer after the announcement, consistent with good borrowers being downgraded into the pool. Table 6 shows strategic upgrades for low-quality multiple-lender borrowers. Table 7 shows rating informativeness changes in the predicted direction.

Economic message. The same numerical rating becomes less informative when disclosure creates strategic reporting incentives.

Table 5
Information sharing and credit rating informativeness

<i>A</i>		
Dependent variable	Default	
Sample lender	Single and rating = 2	
	(1)	(2)
Default horizon	6 Mths	12 Mths
Treated	−0.014 (0.102)	−0.084 (0.122)
Treated × Pre	0.273** (0.109)	0.175* (0.100)
Bank controls	Yes	Yes
Borrower controls	Yes	Yes
Time fixed effects	Yes	Yes
Bank fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
No. observations	768	435
R-sq.	0.34	0.43
<i>B</i>		
Dependent variable	Default	
Sample lender	Single and rating = 1	
	(1)	(2)
Default horizon	6 Mths	12 Mths
Treated	0.001 (0.021)	0.009 (0.034)
Treated × Pre	0.017 (0.030)	0.010 (0.028)
Bank controls	Yes	Yes
Borrower controls	Yes	Yes
Time fixed effects	Yes	Yes
Bank fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
No. observations	7,387	4,238
R-sq.	0.15	0.25

This table presents the difference-in-differences effect of the credit rating in the pre-announcement period on default hazard rates for borrowers with a rating = 2, estimated using OLS. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The dependent variable, default, is an indicator variable equal to one if the borrower

(a) Table 5: single-lender informativeness

Table 6
Strategic upgrades of low-quality borrowers by multiple lenders

Dependent variable	Upgrades from 2 to 1						
Sample lender	Multiple						3-month post
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated	−0.030 (0.022)						
Treated × Interim	0.089* (0.051)	0.152** (0.064)	0.209** (0.071)	0.162* (0.094)	0.222*** (0.111)	0.147** (0.061)	0.180** (0.074)
Treated × Post					0.103 (0.099)		
Treated × Interim × Local				−0.034 (0.130)	−0.041 (0.148)		
Treated × Post × Local					−0.014 (0.125)		
Bank controls	Yes	Yes	Yes	Yes	Yes	No	No
Borrower controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Borrower fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes
Bank-time fixed effects	No	No	No	No	No	Yes	Yes
Industry-time fixed effects	No	No	No	No	No	Yes	Yes
No. observations	2,365	2,365	2,365	2,365	2,365	2,365	1,240
R-sq.	0.11	0.36	0.36	0.36	0.36	0.66	0.59

This table presents the difference-in-differences effect of the registry expansion announcement (interim period) and public information (post-expansion period) on upgrade hazard rates, estimated using OLS. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The dependent variable, upgrade, is from rating = 2 to rating = 1 (best). The sample is all treatment and control borrower-bank relationships with a rating = 1 in the pre-announcement period for borrowers with multiple lending relationships in the pre-announcement period. Columns (1) to (6) use a 12-month post-period horizon, while Column (7) uses a three-month post-period horizon. Local is an indicator variable equal to one if the lender is a local bank. The unit of observation is at the borrower-bank-month level. Bank controls include log(total assets), return on assets, capitalization (%), log(number of branches), nonperforming loans-to-total assets, deposits-to-total assets, and growth on total assets, while borrower controls include the log(debt), state, size, industry classification, and urban. Standard errors are clustered at the borrower level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level.

(b) Table 6: strategic upgrades

Table 7
Informativeness of multiple lenders credit ratings

<i>A</i>		
Dependent variable	Default	
Sample lender	Multiple and rating = 1	
	(1)	(2)
Default horizon	6 Mths	12 Mths
Treated	0.020 (0.015)	0.032 (0.022)
Treated $\times$ Pre	-0.033** (0.016)	-0.025* (0.015)
Bank controls	Yes	Yes
Borrower controls	Yes	Yes
Time fixed effects	Yes	Yes
Bank fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
No. observations	23,539	13,355
R-sq.	0.07	0.11
<i>B</i>		
Dependent variable	Default	
Sample lender	Multiple and rating = 2	
	(1)	(2)
Default horizon	6 Mths	12 Mths
Treated	0.015 (0.046)	0.017 (0.072)
Treated $\times$ Pre	-0.025 (0.066)	-0.070 (0.077)
Bank controls	Yes	Yes
Borrower controls	Yes	Yes
Time fixed effects	Yes	Yes
Bank fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
No. observations	1,700	924
Adj. R-sq.	0.178	0.15

This table presents the difference-in-differences effect of the credit rating in the pre-announcement period on default hazard rates for borrowers with a credit rating of 1, estimated using OLS. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). The dependent variable, Default, is an indicator variable equal to one if the

Table 1: Table 7: multiple-lender rating informativeness

5.5 Tables 8–9: credit-allocation effects

Read: Table 8 shows information sharing improves access for some borrowers, including more foreign-bank lending and more bank relationships. Table 9 shows strategically downgraded borrowers do not obtain the same benefits.

Economic message. Disclosure reduces adverse selection, but manipulation can preserve bank rents and distort credit allocation.

Table 8
Information sharing and changes in bank lending relationships

Sample	Single lender						
	All borrowers				Rating = 1		Rating >1
	log(total debt)	log(#banks)	% local	% foreign	% local	% local	% local
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated × Post	0.021 (0.064)	−0.017 (0.025)	−0.032** (0.015)	0.026* (0.015)	−0.015* (0.009)	−0.017* (0.010)	−0.005 (0.005)
Borrower controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	No	No	Yes	Yes	Yes
Industry-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. observations	11,698	11,698	11,698	11,698	11,698	9,737	1,961
R-sq.	0.57	0.68	0.44	0.24	0.99	0.99	0.99

Sample	Multiple lender						
	All borrowers				Rating = 1		Rating >1
	log(total debt)	log(#banks)	% local	% foreign	% local	% local	% local
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated × Post	0.009 (0.038)	0.044** (0.018)	−0.006 (0.010)	0.009 (0.011)	−0.008 (0.008)	−0.006 (0.008)	−0.002 (0.023)
Borrower controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	No	No	Yes	Yes	Yes
Industry-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. observations	18,838	18,838	18,838	18,838	18,838	16,534	2,304
R-sq.	0.58	0.79	0.411	0.231	0.97	0.97	0.98

This table presents the difference-in-differences effect of public information (post-expansion period) on bank lending-relationships. Treatment (control) borrowers are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). In panels A and B, the sample is all treatment and control borrowers with single and multiple lending relationships in the pre-announcement period, respectively. Dependent variables are the log(total debt), the number of lending relationships (log(#banks)), the fraction of lending relationships that are with local banks (% local), which reflect relationship banks, and the fraction of lending relationships that are with foreign banks (% foreign), which reflect transactional banks. The unit of observation is at the borrower-bank-month level. Columns (1) to (5) include all borrowers; Column (6) includes borrowers that maintain a rating that equals one with all lenders; and Column (7) includes those borrowers with at least one rating greater than one. Borrower controls include the state, size, industry classification, and urban in Column (1) and log(total debt), state, size, industry classification, and urban in Columns (2)-(7). The unit of observation is at borrower-month level. Standard errors are clustered at the borrower level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level.

Table 9
Information manipulation and changes in bank lending relationships

Sample	Single lender			
	Treatment borrowers w/rating = 1 in the pre-period			
	log(total debt)	log(#banks)	% local	% foreign
Dependent variable	(1)	(2)	(3)	(4)
Downgraded × Post	0.055 (0.176)	−0.130* (0.070)	0.021** (0.010)	−0.017* (0.009)
Borrower controls	Yes	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	No	No
Industry-time fixed effects	Yes	Yes	Yes	Yes
No. observations	2,092	2,092	2,092	2,092
R-sq.	0.57	0.77	0.98	0.96

This table presents the difference-in-differences effect of public information (post-expansion period) on bank-relationships for those borrowers that were strategically downgraded. Treatment borrowers are those with total debt between \$150,000 and \$200,000 during the pre-announcement period (January to March 1998). The sample is all treatment borrowers with single lending relationships and a rating that equals one in the pre-announcement period. Dependent variables are the log(total debt), the number of relationships (log(#banks)), the fraction of lending relationships that are with local banks (% local), which reflect relationship banks, and the fraction of lending relationships that are with foreign banks (% foreign). The unit of observation is at the borrower-bank-month level. Borrower controls include the state, size, industry classification, and urban in Column (1) and log(total debt), state, size, industry classification, and urban in Columns (2)-(4). The unit of observation is at borrower-month level. Standard errors are clustered at the borrower level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level.

(a) Table 8: relationship changes

(b) Table 9: manipulated borrowers

6 Short summary

- The paper studies the 1998 expansion of the Argentine public credit registry.
- The reform created a clean interim period between announcement and implementation.
- Banks downgraded high-quality single-lender borrowers before disclosure.
- Banks upgraded low-quality multiple-lender borrowers before disclosure, consistent with avoiding creditor runs.
- The manipulation made ratings less informative after the reform announcement.
- The credit registry still improved credit access for some borrowers, but strategically downgraded borrowers did not benefit as much.
- The central lesson is that disclosure policy must consider the incentives of the agents who produce the disclosed information.

7 Conclusion

7.1 Core conclusion

Giannetti, Liberti, and Sturgess show that public information sharing can change the incentives of the information provider. Banks forced to disclose borrower ratings may manipulate those ratings before disclosure. The result is not simply more information; it is strategically altered information.

7.2 Economic intuition

For good borrowers with one bank, truthful disclosure reduces the incumbent bank's informational monopoly. The bank therefore has an incentive to downgrade the borrower before disclosure. For bad borrowers with several banks, bad news can trigger a creditor run. The bank therefore has an incentive to upgrade the borrower before disclosure.

7.3 Takeaway

Credit registries can reduce adverse selection and improve credit access, but only if the information they disseminate remains reliable. Registry design should focus not only on what information is shared, but also on who supplies it, when it is supplied, and how the supplier's incentives change when disclosure becomes mandatory.